

Epidemic Preparedness and Vaccine Equity: Access for All (142)

**JOHN R BASSLER¹, CASEY L DANIEL²,
DAVID T REDDEN³, SCOTT HARRIS⁴,
PAUL C ERWIN⁵**

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12:15 PM - 12:45 PM ET**

- 1 DRPH CANDIDATE, THE U. OF ALABAMA AT BIRMINGHAM (UAB), DEPARTMENT OF BIostatISTICS**
- 2 ASSOCIATE PROFESSOR, U. OF SOUTH ALABAMA (USA)**
- 3 PROFESSOR, EDWARD VIA COLLEGE OF OSTEOPATHIC MEDICINE (VCOM)**
- 4 STATE HEALTH OFFICER, ALABAMA DEPARTMENT OF PUBLIC HEALTH (ADPH)**
- 5 DEAN, SCHOOL OF PUBLIC HEALTH, UAB**

Background and Methods



What's The Problem

- COVID-19 vaccine uptake in Alabama (~**52%** fully vaccinated, as of May 2023; **compared to ~70% nationally** ¹)

What's Possible?

- **Dollar General and Rural America** ²
- **“Shots at the Shop”** - COVID-19 vaccine initiatives included **Black-owned barbershops and hair salons** as sources for accurate information and **vaccine administration** ³

Would It Work?

- **County-level** vaccine rates (fully vaccinated)
- All Dollar General (DG) locations
- All COVID-19 vaccine providers ⁴ (including Drive Thru and Walk In Clinics)
- **DG-to-Vaccine Provider Ratio**

Now What?

- This analysis DOES NOT imply that $\uparrow$ DG = $\downarrow$ Vaccines
- Counties with low COVID-19 vaccine rates have more access to DGs than traditional pharmacies (**DG / Vaccine Provider Ratio**)
- **Can DG locations in pharmacy deserts can be leveraged to increase vaccine access?**

Is It Worth It?

- Herd Immunity
- Prevention: **Vaccine barriers = Treatment barriers**
- There is **no such thing as isolation**
- The *next* epidemic

1. USAFacts. US Coronavirus vaccine tracker. USAFacts. Published July 17, 2025. Accessed July 17, 2025. <https://usafacts.org/visualizations/covid-vaccine-tracker-states/>

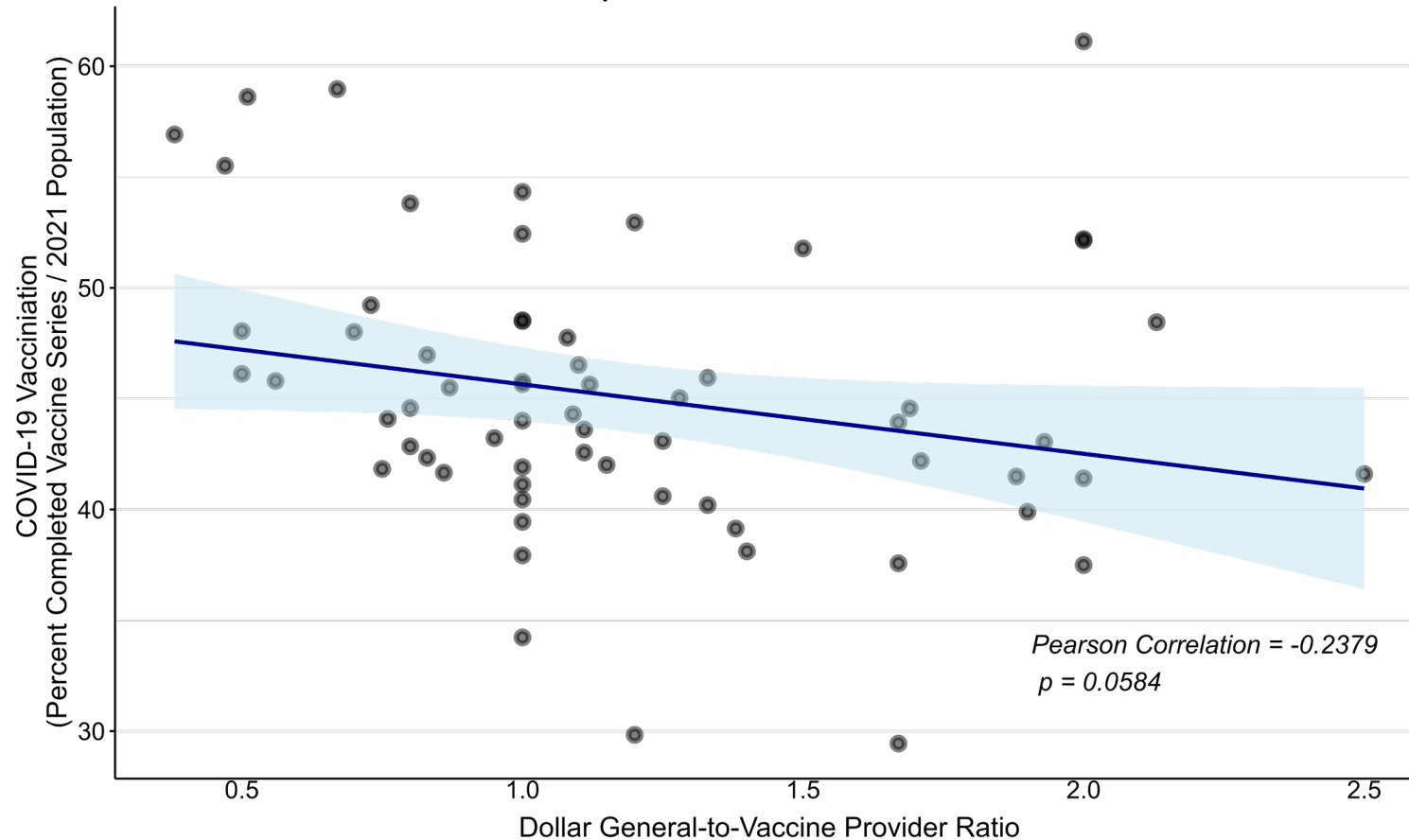
2. How Dollar General Became Rural America's Store of Choice - TREK Investment Group. TREK Investment Group. Published December 7, 2017. Accessed July 17, 2025. <https://www.trekig.com/in-the-press/how-dollar-general-became-rural-americas-store-of-choice/>

3. COVID-19 Vaccine Initiatives included Black-owned barbershops and hair salons as sources for accurate information and administer vaccines

4. Vaccines.gov: COVID-19 vaccinating provider locations. Cdc.gov. Published April 28, 2021. Accessed December 8, 2024. https://data.cdc.gov/Vaccinations/Vaccines-gov-COVID-19-vaccinating-provider-locatio/5jp2-pgaw/about_data; and, The Alabama COVID-19 Vaccine Clinic Dashboard, <https://www.arcgis.com/apps/dashboards/2b4627aa70c5450791a7cf439ed047ec>

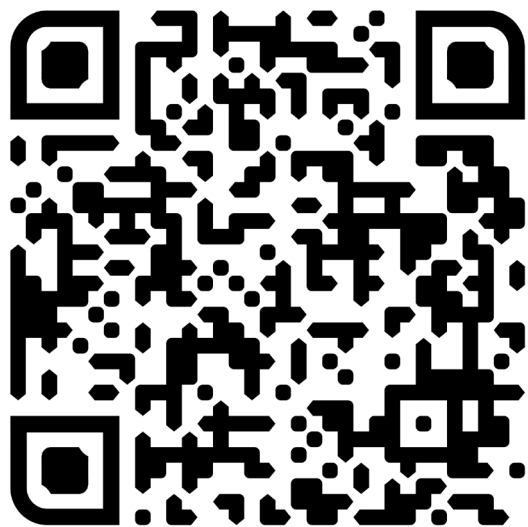
Results

Figure 1. Linear trend (95% Confidence Interval) between Alabama County COVID-19 vaccination rates and ratio of Dollar General locations to COVID-19 vaccination providers



- **Proof of concept** (General Linear Models used to assess associations with COVID-19 vaccine rates)
- SDI Components, increases in **lack of education** ($p = .002$) and **unemployment** ($p = .005$), associated with **lower COVID-19 vaccine rates**
- **Significance is not the point**, and there is no such thing as “approaching” significance! **DG-to-Vaccine Provider Ratio** ($p = .058$)
- The trend is still informative
- Counties with **higher ratios of DGs to vaccine providers** were found to have **lower rates of COVID-19 vaccination**

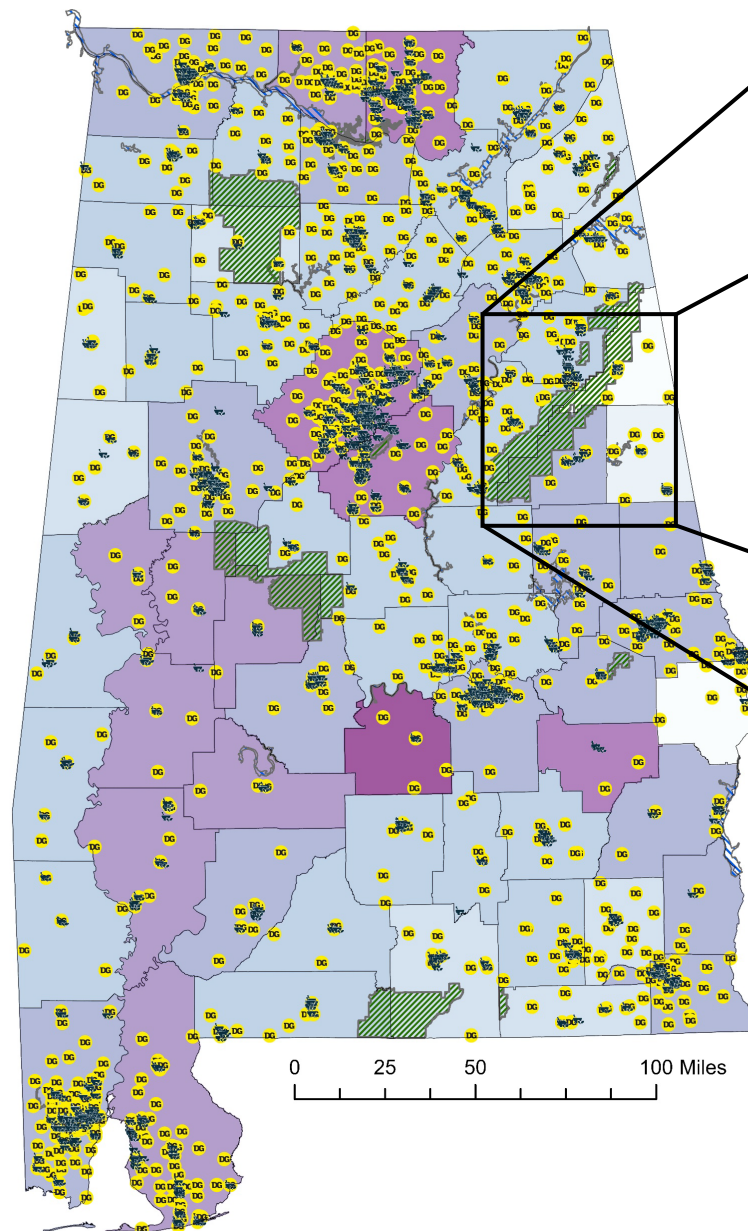
Results



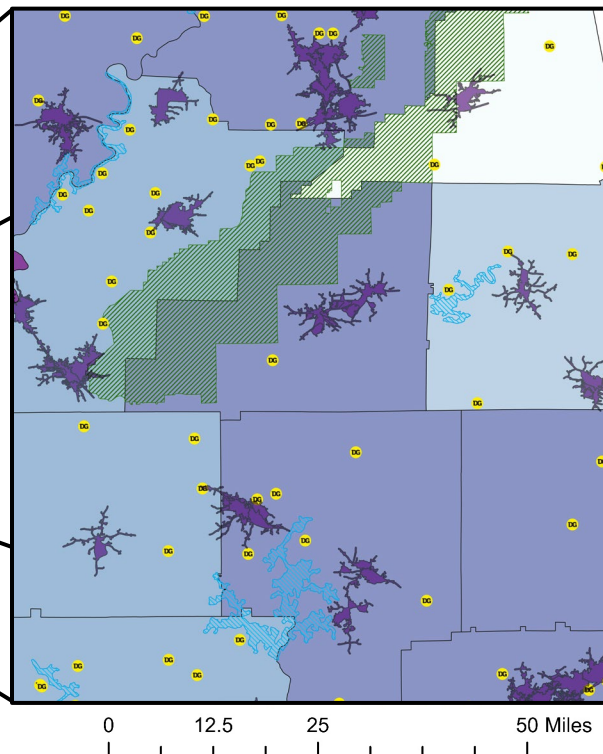
<https://jbassler.shinyapps.io/AL-COVID19-DG/>



Figure: County COVID-19 Vaccination %, overlaid with DGs and vaccine providers



Inset: County COVID-19 Vaccination %, overlaid with DGs and 5-mile vaccine provider isochrones



COVID-19 Vaccination (County Pop. %)

- < 30%
- 30% - 34%
- 35% - 39%
- 40% - 44%
- 45% - 49%
- 50% - 54%
- 55% - 59%
- ≥ 60%

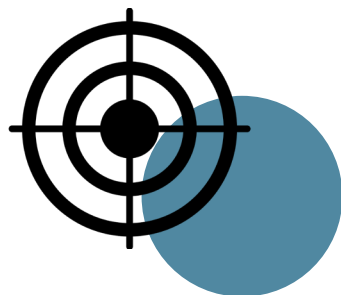
COVID-19 Vaccine Providers

Dollar General Locations

National and State Parks

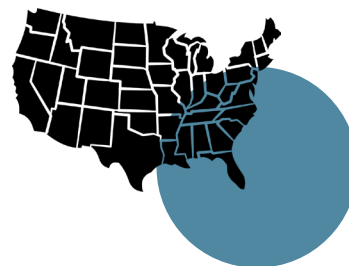
Lakes

Conclusions and Next Steps



VACCINE EQUITY

- In areas with historically low vaccine rates and/or limited access to vaccine providers, local DGs could provide a stable distribution infrastructure, and point-of-contact to expand vaccine delivery



EPIDEMIC RESPONSE

- Similar to “Shots at the Shop,” during a public health crisis, DGs could be utilized to reach at-risk populations and improve access in vaccination deserts



ROUTINE VACCINATIONS

- Strategic co-location of mobile vaccination initiatives at DGs in vulnerable communities can provide a scalable method to reduce geographical vaccination disparities⁵